Pham Thanh Tan

DevOps & Virtualization Engineer

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Professional Summary

DevOps & Virtualization Engineer with hands-on experience deploying and operating on-premise and containerized infrastructure. Specialized in VMware vSphere (vCenter, ESXi, vSAN, Horizon), Docker, Kubernetes, and system automation. Combines a traditional-virtualization mindset with cloud-native DevOps practice to improve stability, performance, and scalability of enterprise environments.

Core Competencies

Virtualization & Infrastructure: VMware vSphere (ESXi, vCenter), vSAN, VMware Horizon (VDI), Dell PowerEdge, HP ProLiant

Containers & DevOps: Docker, Kubernetes, Portainer, Traefik, CI/CD fundamentals, automation

Networking & Security: Cisco & HP switching, VLAN/L2/L3, MPLS, HAProxy, Keepalived, pfSense, OPNsense, FortiGate, Sophos

Operating Systems & Scripting: Linux (Ubuntu, Debian), Windows Server, Python, Bash, PowerShell

AI Infrastructure: LLM inference serving (Aphrodite-engine, vLLM, SGLang), OpenAI-compatible APIs, GPUStack, CUDA 12.x and legacy-GPU tuning, FastAPI

Work Experience

System Engineer, Huong Nghiep A Au – Ho Chi Minh City, Vietnam — Dec 2022 – present

- Operate the on-premise virtualization platform on VMware vSphere 7.0 — a 4+ host ESXi cluster plus standalone hosts supporting more than 25 virtual machines.
- Engineered and deployed a container platform with Docker and Docker Compose, automating deployment and lifecycle management through Portainer and Traefik, cutting manual configuration time.
- Administered central NAS storage and implemented a 3-2-1 backup and recovery strategy (local, cloud, off-site) to protect data integrity.
- Deployed and administered VMware Horizon (VDI) with vGPU to serve dedicated AI workloads (vLLM, SGLang).
- Tuned vSAN storage and VMware Distributed Switch networking for performance and high availability.
- Kept .NET applications stable across a mixed Windows Server, Ubuntu, and Debian estate.
- Technologies: VMware vSphere 7.0 (ESXi, vCenter), vSAN, Horizon, Docker, Portainer, Traefik, NAS, Dell PowerEdge, HP ProLiant.

Network Engineer Intern, Huong Nghiep A Au – Ho Chi Minh City, Vietnam — Aug 2022 – Dec 2022

- Researched and proposed IT solutions for collaboration (OwnCloud/OnlyOffice) and an on-premise SharePoint 2013 deployment.
- Independently deployed open-source firewalls (pfSense, OPNsense) with High Availability, anti-virus proxy (Squid + ClamAV), and DDNS.
- Configured Cisco and HP switches — VLANs and routing for branch offices across an MPLS network — to enforce network segmentation.

IT Helpdesk, FPT University – Ho Chi Minh City, Vietnam — Mar 2020 – July 2022

- Delivered Tier 1 support for hardware, software, and network incidents for students and staff.
- Analyzed recurring incidents and documented standardized troubleshooting procedures, improving support efficiency and reducing time to resolution.
- Maintained and repaired campus network and security equipment (Cisco, Grandstream, Draytek, CCTV).

Projects

SDN Research on Kubernetes — Capstone – FPT University — Sept 2023 – Dec 2023

Led a 5-person team to design and build a complete virtualized data-center lab on EVE-NG, then deploy a highly available Kubernetes cluster on top of it.

- Deployed a 6-node High Availability Kubernetes cluster (2 control plane, 4 worker) with HAProxy and Keepalived for control-plane redundancy.
- Developed Python scripts that generate NetworkPolicy templates, standardizing and accelerating pod security configuration (deny-by-default, whitelist, cross-namespace isolation).
- Result: 8.4/10 — recognized for virtualization design depth and practical container orchestration.
- Project documentation

Aphrodite LLM Deployment for Pascal GPUs 2025

- Built a Docker image of a patched Aphrodite Engine on CUDA 12.1 to serve modern LLMs on legacy NVIDIA Pascal cards (Tesla P40, 24 GB), a platform the upstream stack no longer targets.
- Automated model/runtime configuration with a Python configurator, plus half-precision and enforce-eager settings that fit a 24 GB card and expose an OpenAI-compatible endpoint.

Kokoro Vietnamese TTS Server 2025

- Packaged the Kokoro-Vietnamese 82M model into a FastAPI service that is a drop-in replacement for the OpenAI POST /v1/audio/speech endpoint, with SSE and raw-audio streaming.
- Added 14 Vietnamese voicepacks with runtime swapping on a shared model backbone, a queue scheduler with backpressure, health/metrics probes, and GPUStack deployment manifests.

TapViet Studio 2025

- Shipped a browser-only Vietnamese handwriting-worksheet generator (JavaScript, HTML, CSS) that renders Ministry-of-Education standard glyph grids and exports print-ready PDF.

Education

FPT University, Bachelor in Information Assurance – Ho Chi Minh City, Vietnam 2019 – 2023

- GPA: 7.38/10

Certifications

Enterprise DevOps Practices (Kubernetes, Logging, Monitoring): DevOpsEdu.vn — 03/2023

Cisco Certified Network Associate (CCNA 200-301): VN Pro — 03/2022

CertNexus Certified Ethical Emerging Technologist: FPT Coursera — 03/2022

CyberOps Associate: Cisco NetAcad — 11/2021

CCNAv7 — Introduction to Networks: Cisco NetAcad — 07/2021

Activities

Vice President, FPT Information Assurance Club Sept 2020 – Dec 2023

- Coordinated technical workshops and events with the leadership team.
- Contributed to strategic planning for a knowledge-sharing community.

Volunteer, Ho Chi Minh Communist Youth Union Jan 2019 – Dec 2021

- Supported community volunteer activities with fellow union members.

Interests

Exploring new technologies and gadgets

Collaborating on team activities and social events

Traveling